Block practice · Differentiation and its applications

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Connected rates: linking two changing quantities

Differentiating twice and proving a relation between y and its derivatives

Finding where a curve has a given gradient

OLevels.com SCHOOL

Practice workbook · Connected rates: linking two changing quantities

Using a known rate of one quantity (for example volume) to find the rate of another (for example surface area) through a shared variable.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-CAL-15

1. Question 9 · part whole · 7 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part whole only.

9 A cube has edges of length x cm.

The total surface area, A cm^2, of the cube is increasing at a constant rate of 0.45 cm^2/s

Find the rate of increase, in cm^3/s, of the volume of the cube at the instant when the total surface area of the cube is 384 cm^2

(7)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 9 · part whole · 7 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part whole only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

(Total for Question 9 is 7 marks)

1. Question 9 · part whole · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 10 · part c · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

10

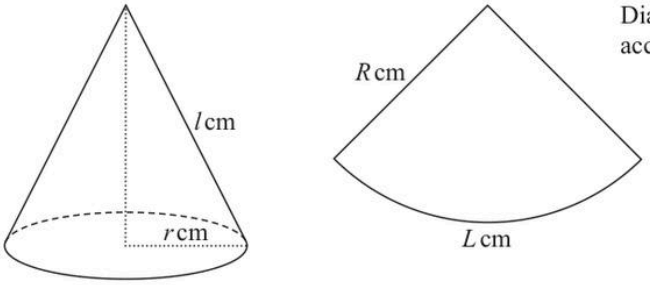


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right circular cone with base radius r cm and slant height l cm.
Figure 4 also shows a sector of a circle with radius R cm and arc length L cm.

The area of the curved surface of the cone is A cm^2

By considering how the sector of the circle can be folded to exactly form the curved surface of the cone with R and L suitably chosen,

(a) prove that $A = \pi rl$

(4)

Sand is poured onto a horizontal surface at a constant rate of 1.5 cm^3/s.
The sand forms a pile in the shape of a right circular cone with its base on the surface.
The curved surface area of the cone, A cm^2, increases in such a way that the height of the cone is always three times the radius of the base of the cone.

Given that $\frac{dA}{dr} = k\pi r$, where k is a constant,

(b) find the exact value of k.

(3)

(c) Calculate the rate, in cm^2/s, to 3 significant figures, at which the curved surface area of the pile is increasing when the height of the pile is 24 cm.

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

26

P 6 2 2 8 0 A 0 2 6 3 2

2. Question 10 · part c · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 10 · part c · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 10 · part c · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

(Total for Question 10 is 12 marks)

2. Question 10 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 6 · part b · 8 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

6

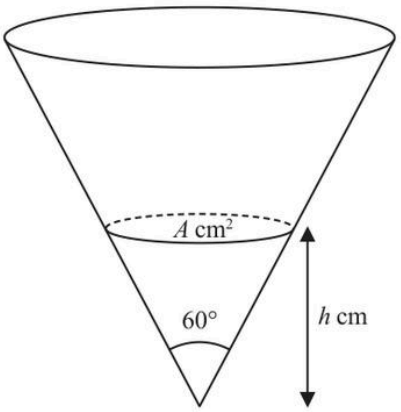


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a container in the shape of a right circular cone.
The container is fixed with its axis of symmetry vertical.
The vertical angle of the container is 60° as shown in the diagram.

At time t seconds, $t > 0$, the height of oil in the container is h cm and the volume of oil in the container is V cm^3

(a) Show that $V = \frac{1}{9}\pi h^3$

(3)

At time t seconds the surface area of oil in the container is A cm^2, as shown in Figure 4

Oil is dripping out of the bottom of the container at a constant rate of 4 cm^3/s.

(b) Find the exact rate of change, in cm^2/s, of the surface area of oil in the container when $h = 24$

(8)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 4 0 9 8 A 0 1 6 3 2

3. Question 6 · part b · 8 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 6 · part b · 8 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 6 · part b · 8 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 6 continued

OLevels.com SCHOOL

(Total for Question 6 is 11 marks)

3. Question 6 · part b · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 9 · part b · 8 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

9

Diagram **NOT** accurately drawn

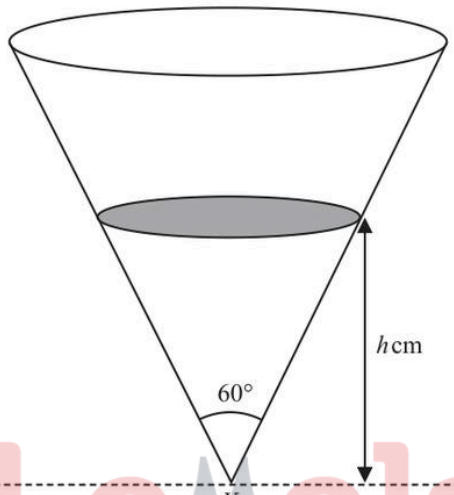


Figure 2

Figure 2 shows a hollow right circular cone fixed with its axis of symmetry vertical.

The vertical angle of the cone is 60°
Initially the cone is empty.

At time $t = 0$ liquid starts to fill the cone at a constant rate of $0.2\,\text{cm}^3/\text{s}$
At time t seconds after the liquid starts to fill the cone, the height of the liquid is h cm above X

(a) Show that $h = \sqrt[3]{\frac{9t}{5\pi}}$

(5)

The surface area of the liquid, shown shaded in Figure 2, is increasing at a constant rate of $p\,\text{cm}^2/\text{s}$ when $t = 6$

(b) Find, to 3 significant figures, the value of p

(8)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

26

P 7 8 9 6 4 A 0 2 6 3 6

4. Question 9 · part b · 8 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 9 · part b · 8 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 9 · part b · 8 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 9 continued

OLevels.com SCHOOL

(Total for Question 9 is 13 marks)

4. Question 9 · part b · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 9 · part b · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

9

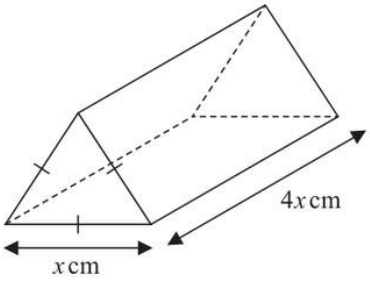


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a metal solid S.

The solid is a right triangular prism.

The cross section of S is an equilateral triangle with sides of length x cm.
The length of S is $4x$ cm.

The prism is being heated so that the cross sectional area is increasing at a constant rate of $0.03\,\text{cm}^2/\text{s}$.

(a) Find, giving your answer to 3 significant figures, $\frac{dx}{dt}$ when $x = 2$ **(5)**

(b) Find the rate of increase, in cm^3/s, of the volume of S when $x = 2$ **(3)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5. Question 9 · part b · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 9 · part b · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 9 · part b · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.



Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 9 is 8 marks)

5. Question 9 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 10 · part whole · 9 marks

2020 JAN · Paper 02

Further practice · Attempt part whole only.

10 The volume of a sphere is increasing at a constant rate of $40\,cm^3/s$.

Find the rate of increase, in cm^2/s, of the surface area of the sphere at the instant when the radius is 4 cm.

(9)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 10 · part whole · 9 marks

2020 JAN · Paper 02

Further practice · Attempt part whole only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

(Total for Question 10 is 9 marks)

6. Question 10 · part whole · 9 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 7 · part b · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

7 The length of each side of a cube S_1 is increasing at a constant rate of 0.1 m/s.

(a) Find, in m^3/s, the rate of increase of the volume of the cube S_1 when the length of each side of the cube is 2 m.

(4)

The total surface area of a different cube S_2 is increasing at a constant rate of $0.05\,m^2/s$.

(b) Find in m^3/s, the rate of increase of the volume of the cube S_2 when the length of each side of the cube is 6 m.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

20

P 6 2 2 8 3 A 0 2 0 4 0

7. Question 7 · part b · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 7 continued

OLevels.com SCHOOL

7. Question 7 · part b · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

22

P 6 2 2 8 3 A 0 2 2 4 0

7. Question 7 · part b · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

(Total for Question 7 is 9 marks)

7. Question 7 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Differentiating twice and proving a relation between y and its derivatives

Differentiating twice (usually with the product rule) and substituting y, dy/dx and d2y/dx2 into a given relation to show it holds or to find constants.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-CAL-03

1. Question 8 · part whole · 8 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part whole only.

8 Given that $y = e^{3x} \sin 2x$

show that $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 13y = 0$

(8)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 5 9 9 3 9 A 0 2 4 3 6

1. Question 8 · part whole · 8 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part whole only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 8 · part whole · 8 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part whole only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

26

P 5 9 9 3 9 A 0 2 6 3 6

1. Question 8 · part whole · 8 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part whole only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 8 is 8 marks)

1. Question 8 · part whole · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 9 · part whole · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

9 Given that

$$x = e^{-t} \sin 2t$$

show that

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = 0$$

(8)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 9 · part whole · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 9 · part whole · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 9 · part whole · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 9 is 8 marks)

2. Question 9 · part whole · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 3 · part whole · 8 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

3 Given that $y = e^{3x} \sin 2x$

show that $13y + \frac{d^2y}{dx^2} = 6\frac{dy}{dx}$

(8)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 3 · part whole · 8 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 3 is 8 marks)

3. Question 3 · part whole · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 6 · part a.ii · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part a.ii only.

6 (a) Given that $y = (4x - 3)e^{2x}$

(i) find $\frac{dy}{dx}$ (3)

(ii) show that $(4x - 3)\frac{dy}{dx} = (8x - 2)y$ (2)

(b) Differentiate $\frac{\sin 5x}{(x-3)^2}$ with respect to x (3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

4. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 7 · part b · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

7 $y = e^{2x} \cos 2x$

(a) Show that

$$\frac{dy}{dx} = 2y - 2e^{2x} \sin 2x$$

(4)

(b) Hence show that

$$\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$$

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

5. Question 7 · part b · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 7 · part b · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 7 · part b · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

(Total for Question 7 is 9 marks)

5. Question 7 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 9 · part b · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

9

$$y = e^{-4t}\cos 2t$$

(a) Show that $2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$ (3)

Given that $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = 0$ where M and N are integers

(b) find the value of M and the value of N (5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

6. Question 9 · part b · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 9 continued

OLevels.com SCHOOL

6. Question 9 · part b · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 9 · part b · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 9 continued

OLevels.com SCHOOL

(Total for Question 9 is 8 marks)

6. Question 9 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 6 · part whole · 8 marks

2025 NOV · Paper 01

Further practice · Attempt part whole only.

6 $y = e^{4x} \cos 3x$

Show that

$$\frac{d^2y}{dx^2} + Ay = B\frac{dy}{dx}$$ where A and B are integers to be found. **(8)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 6 · part whole · 8 marks

2025 NOV · Paper 01

Further practice · Attempt part whole only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 6 is 8 marks)

7. Question 6 · part whole · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 5 · part whole · 7 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

5

$$y = e^{2x}(x^2 - 5x)$$

Show that $2e^{2x} = \frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y$ (7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 6 5 0 8 A 0 1 0 3 2

8. Question 5 · part whole · 7 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 5 continued

OLevels.com SCHOOL

(Total for Question 5 is 7 marks)

8. Question 5 · part whole · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Finding where a curve has a given gradient

Setting dy/dx equal to a known gradient (from a parallel line or a normal) to find the points, then using them for tangent constants, lines or lengths.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-CAL-06

1. Question 10 · part e · 7 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part e only.

10 A curve C has equation

$$y = \frac{7-x}{2x+3} \qquad x \neq -\frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis

(ii) parallel to the x-axis

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

C passes through the point A with coordinates $\left(\frac{1}{2}, \frac{13}{8}\right)$

(d) Show that the gradient of C at A is $-\frac{17}{16}$

(3)

C also passes through the point P

Given that the tangent to C at P is parallel to the tangent to C at A

(e) find an equation of the tangent to C at P
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(7)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

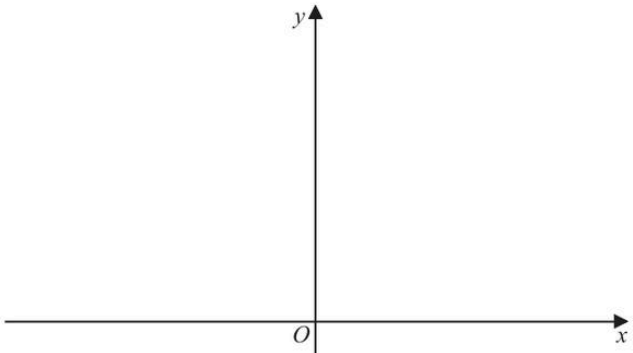
OLevels.com — SCHOOL —

1. Question 10 · part e · 7 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part e only.

Question 10 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 10 · part e · 7 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part e only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

28

P 7 2 8 6 7 A 0 2 8 3 2

1. Question 10 · part e · 7 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part e only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

(Total for Question 10 is 17 marks)

1. Question 10 · part e · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 10 · part f · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part f only.

10 The curve C has equation

$$y = \frac{5x-2}{ax+b} \qquad x \neq -\frac{b}{a}$$

The asymptote to C that is parallel to the y-axis has equation $x = -\frac{4}{3}$

C crosses the y-axis at the point with coordinates $\left(0, -\frac{1}{4}\right)$

(a) (i) Show that $b = 8$

(ii) Find the value of a

(3)

(b) Write down the equation of the asymptote to C that is parallel to the x-axis.

(1)

(c) Find the coordinates of the point where C crosses the x-axis.

(1)

(d) Using calculus, show that at every point on C, the gradient is positive.

(4)

(e) Using the axes on the next page, sketch C
Clearly label the asymptotes and the coordinates where C crosses the coordinate axes.

(3)

The gradient of C is $\frac{13}{100}$ at the point E and at the point F

(f) Find the length of the line EF, giving your answer to 2 decimal places.

(6)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

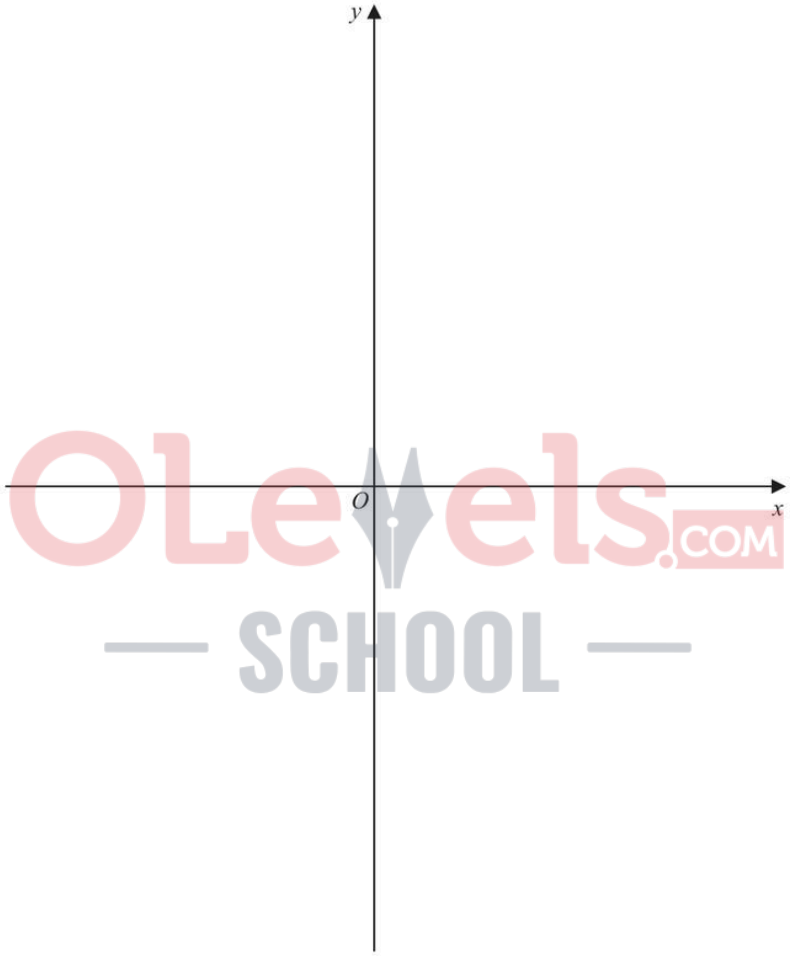
P 7 6 5 0 9 A 0 2 6 3 2

2. Question 10 · part f · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part f only.

Question 10 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 10 · part f · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part f only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 10 · part f · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part f only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 10 continued

OLevels.com SCHOOL

(Total for Question 10 is 18 marks)

2. Question 10 · part f · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 10 · part d.i · 0 marks

2022 JAN · Paper 02

Further practice · Attempt part d.i only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 10 · part d.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 10 · part d.ii · 8 marks

2022 JAN · Paper 02

Further practice · Attempt part d.ii only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 10 · part d.ii · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 10 · part e.i · 0 marks

2022 JAN · Paper 02

Further practice · Attempt part e.i only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

5. Question 10 · part e.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 10 · part e.ii · 3 marks

2022 JAN · Paper 02

Further practice · Attempt part e.ii only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

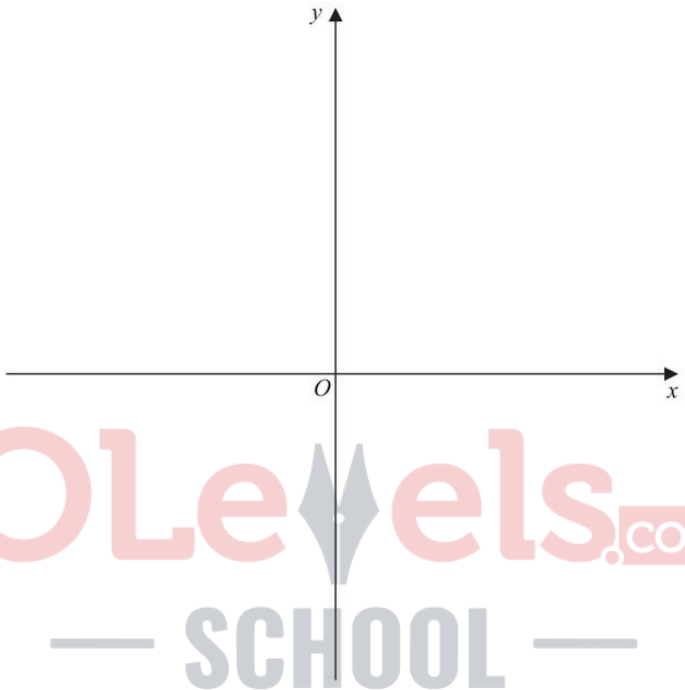
OLevels.com SCHOOL

6. Question 10 · part e.ii · 3 marks

2022 JAN · Paper 02

Further practice · Attempt part e.ii only.

Question 10 continued



OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Turn over for spare axes if you need to redraw your sketch.

6. Question 10 · part e.ii · 3 marks

2022 JAN · Paper 02

Further practice · Attempt part e.ii only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

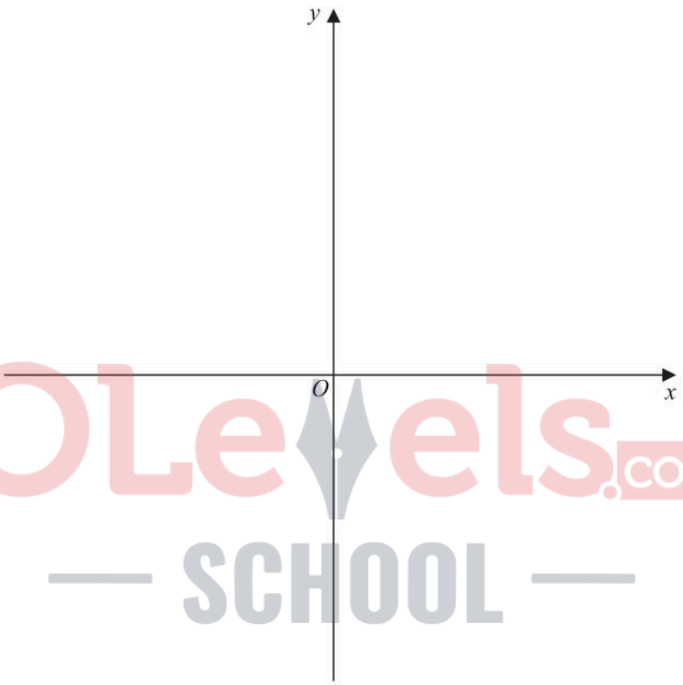
6. Question 10 · part e.ii · 3 marks

2022 JAN · Paper 02

Further practice · Attempt part e.ii only.

Question 10 continued

Only use these axes if you need to redraw your sketch.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

6. Question 10 · part e.ii · 3 marks

2022 JAN · Paper 02

Further practice · Attempt part e.ii only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 10 is 18 marks)

6. Question 10 · part e.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 10 · part e · 7 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

10 A curve C has equation

$$y = \frac{7x - 2}{2x - 3} \qquad x \neq \frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis,

(ii) parallel to the x-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using calculus, show that at every point on the curve, the gradient of C is negative.

(4)

(d) Using the axes on the opposite page, sketch C.
Show clearly and label with their equation any asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The straight line l is the normal to C at the point A.
The x coordinate of A is positive and the gradient of l is 17
The line l also intersects C at the point B.

(e) Find the exact coordinates of B.

(7)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

30

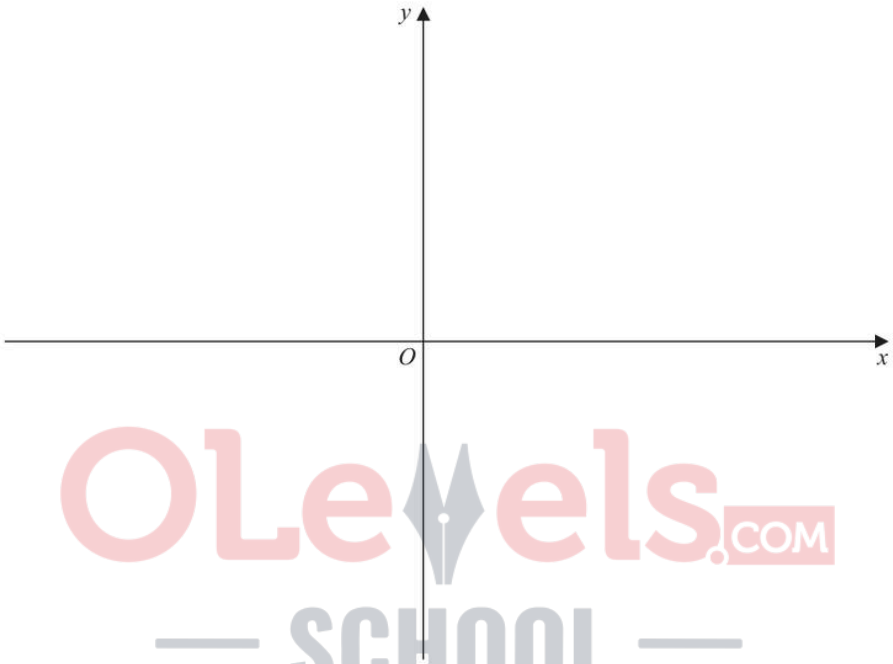
P 7 1 6 4 2 A 0 3 0 3 6

7. Question 10 · part e · 7 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 10 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 10 · part e · 7 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 10 continued

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 10 · part e · 7 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.



Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 10 is 18 marks)

7. Question 10 · part e · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 9 · part e · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

9 A curve C has equation $y = \dfrac{3 - 2x}{x + 6}$ where $x \neq -6$

(a) Write down an equation of the asymptote to C that is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Using the axes opposite, sketch the graph of C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

(d) Show that the gradient of the tangent to C is always negative.

(3)

A tangent to C has equation $y = -\dfrac{3}{5}x + k$ where $k > 0$

(e) Find the value of k

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

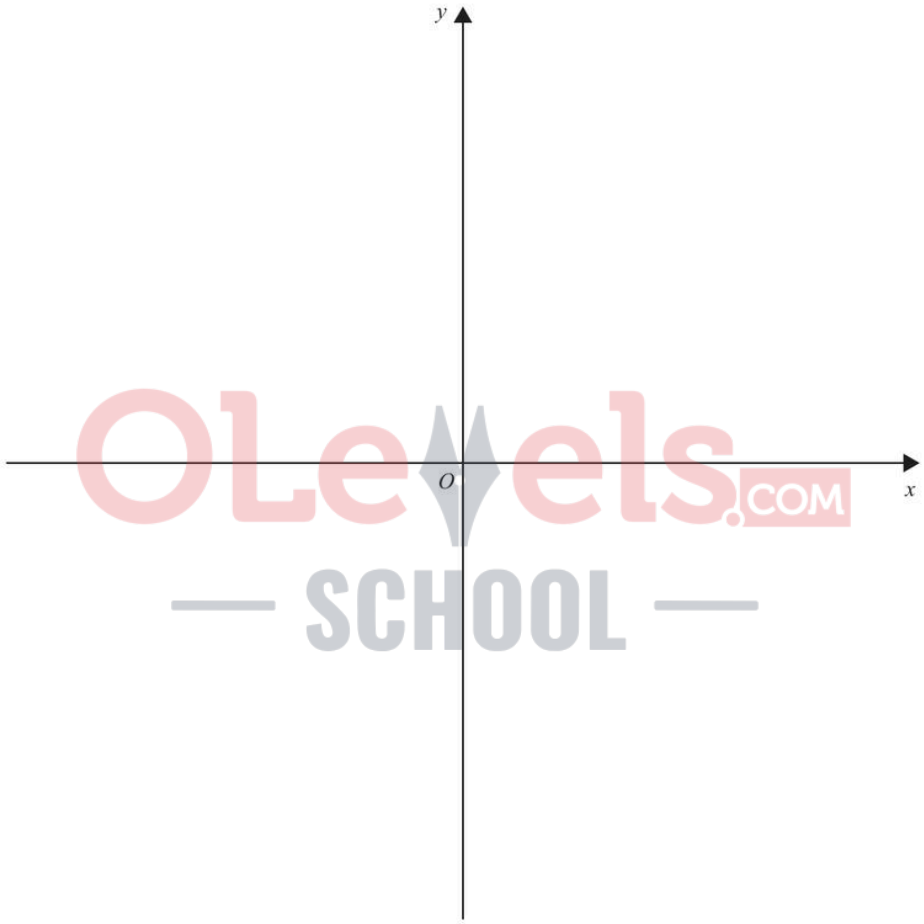
OLevels.COM — SCHOOL —

8. Question 9 · part e · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 9 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

8. Question 9 · part e · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 9 · part e · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 9 is 15 marks)

8. Question 9 · part e · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 10 · part d · 11 marks

2024 NOV · Paper 02

Further practice · Attempt part d only.

10 A curve C has equation

$$y = \frac{5x-2}{3x+2} \qquad x \neq -\frac{2}{3}$$

(a) Find the coordinates of the point where C intersects the

(i) x-axis
(ii) y-axis

(2)

(b) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis
(ii) parallel to the y-axis

(2)

(c) Sketch C on the opposite page.
Show and label the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

Point A lies on C such that the gradient of C at A is parallel to the line with equation $4y - x = 7$

The normal to C at A intersects the x-axis at point D and the y-axis at point E
Given that the x coordinate of A is positive,

(d) find, in its simplified form, the exact length of line DE

(11)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

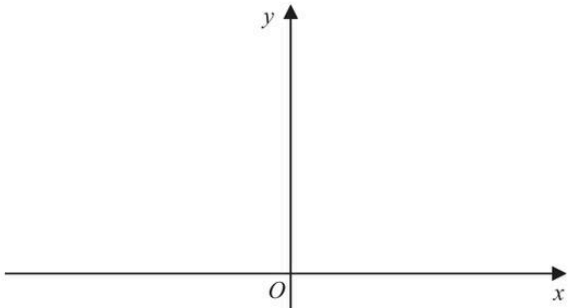
OLevels.COM SCHOOL

9. Question 10 · part d · 11 marks

2024 NOV · Paper 02

Further practice · Attempt part d only.

Question 10 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

9. Question 10 · part d · 11 marks

2024 NOV · Paper 02

Further practice · Attempt part d only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

9. Question 10 · part d · 11 marks

2024 NOV · Paper 02

Further practice · Attempt part d only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

9. Question 10 · part d · 11 marks

2024 NOV · Paper 02

Further practice · Attempt part d only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 10 is 18 marks)

9. Question 10 · part d · 11 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 8 · part a · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

8 $f(x) = 2x^2 - 4x + 9$

The curve C has equation $y = f(x)$
The line l with equation $y - 8x + 9 = 0$ is a tangent to C
The line k is perpendicular to l and is also the tangent to C at the point where $x = p$

(a) Find the value of p

(5)

(b) Show that

(i) $(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$

(ii) $\alpha^4 + \beta^4 = \left((\alpha + \beta)^2 - 2\alpha\beta\right)^2 - 2(\alpha\beta)^2$

(2)

The quadratic equation $f(x) = 0$ has roots α and β
Without solving the equation and using your results from part (b)

(c) form a quadratic equation with integer coefficients, that has roots

$\alpha^3 - \beta$ and $\beta^3 - \alpha$

(8)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 8 · part a · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL